AA 256 Design of Composite Structures

Homework #2

Due date: Friday October 16th, 5 p.m.

1) Show that the plane stress and plane strain stiffness matrices for steel from HW 1, Question 3 are independent of coordinate rotation with respect to the z-axis (out-of-plane). (10 points)

2) Determine the tensile modulus E_1 of a unidirectional layer of T800/3900-2 with an orientation of 20° to the horizontal (0°) axis for: (10 points)

a. A standard coupon of 12 in. length. You may assume that the only nonzero stress in the middle of the coupon is the tensile stress.

b. A short coupon of 1 in. length, fully clamped on both ends. You may assume that the only nonzero strain in the middle of the coupon is the tensile strain.



NOTE: Please use the T800/3900-2 material properties below for Question 2.

Material Properties	**T800/3900-2**
E_{xx} (msi)	23.2
$E_{\gamma\gamma}$ (msi)	1.30
$\nu_{x\gamma}$	0.28
$G_{x\gamma}$ (msi)	0.90
X_t (ksi)	412
X□ (ksi)	225
Y_t (ksi)	10
Y□ (ksi)	24.3
S (ksi)	10
h (in)	0.007

Submission: Upload your calculations, procedure, and results as one PDF to Homework 2 - F26.

1) Show that the plane stress and plane strain stiffness matrices for steel from HW 1, Question 3 are independent of coordinate rotation with respect to the z-axis (out-of-plane). (10 points)

Plane stress

$\sigma_3 = \tau_4 = \sigma_5 = 0$

$$\begin{Bmatrix} \sigma_1' \\ \sigma_2' \\ \sigma_3' \\ \sigma_4' \\ \sigma_5' \\ \sigma_6' \end{Bmatrix} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & 0 & 0 & 0 & 2\cos\theta\sin\theta \\ \sin^2\theta & \cos^2\theta & 0 & 0 & 0 & -2\cos\theta\sin\theta \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos\theta & -\sin\theta & 0 \\ 0 & 0 & 0 & \sin\theta & \cos\theta & 0 \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & 0 & 0 & 0 & \cos^2\theta - \sin^2\theta \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{Bmatrix}$$

$\sigma_3' = \sigma_3$ (0) $\quad \sigma_4' = \sigma_4\cos\theta - \sigma_5\sin\theta$ (0) $\quad \sigma_5' = \sigma_4\sin\theta + \sigma_5\cos\theta$ (0)

$$\therefore T = \begin{bmatrix} \cos^2\theta & \sin^2\theta & 2\cos\theta\sin\theta \\ \sin^2\theta & \cos^2\theta & -2\cos\theta\sin\theta \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

$\{\sigma\} = [C]\{\varepsilon\} \quad \{\sigma'\} = [C']\{\varepsilon'\}$

$\{\sigma'\} = [T]\{\sigma\}$

$\{\varepsilon'\} = [T]\{\varepsilon\}$

$\{\varepsilon\} = [R]\{\varepsilon'\}$

$$\{\sigma'\} = [T][C][R][T]^{-1}[R]^{-1}\{\varepsilon'\}$$

C^{-1}

$$[C'] = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix}$$

$$[R] = \begin{bmatrix} 1 & & \\ & 1 & 0 \\ & 0 & 2 \end{bmatrix} \rightarrow [R] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad [R]^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

$\sigma_3 = 0$, $\tau_4 = 0$, $\sigma_6 = 0$

MATLAB syms:

```
Plane stress C - C_prime diff:
[0, 0, 0]
[0, 0, 0]
[0, 0, 0]
```

Plane strain

$$[C] = \frac{E}{(1+\nu)(1-2\nu)}\begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix}$$

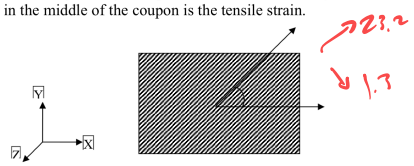
Same MATLAB syms:

```
Plane strain C - C_prime diff:
[0, 0, 0]
[0, 0, 0]
[0, 0, 0]
```

2) Determine the tensile modulus E_1 of a unidirectional layer of T800/3900-2 with an orientation of 20° to the horizontal (0°) axis for: (10 points)

a. A standard coupon of 12 in. length. You may assume that the only nonzero stress in the middle of the coupon is the tensile stress.

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Y_t (ksi)	10
Y_c (ksi)	24.3
S (ksi)	10
h (in)	0.007

Ex Ey ν G

plane stress

$$\{\varepsilon\} = [S]\{\sigma\}$$

$$S = \begin{bmatrix} 1/E_x & -\nu_{yx}/E_y & 0 \\ -\nu_{xy}/E_x & 1/E_y & 0 \\ 0 & 0 & 1/G_{xy} \end{bmatrix}$$

$$\begin{Bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_6 \end{Bmatrix} = \begin{bmatrix} 1/E_x & -\nu_{yx}/E_y & 0 \\ -\nu_{xy}/E_x & 1/E_y & 0 \\ 0 & 0 & 1/G_{xy} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_6 \end{Bmatrix}$$

$$S = \begin{bmatrix} 1/23.2 & -0.28/23.2 & 0 \\ -0.28/23.2 & 1/1.3 & 0 \\ 0 & 0 & 1/0.9 \end{bmatrix} \left[\frac{1}{\text{MPa}}\right] =$$

```
S [1/MPa]:
     0.043103    -0.012069           0
    -0.012069     0.76923            0
            0           0       1.1111
```

w/ $\theta = 20°$

$$\rightarrow T = \begin{bmatrix} \cos^2\theta & \sin^2\theta & 2\cos\theta\sin\theta \\ \sin^2\theta & \cos^2\theta & -2\cos\theta\sin\theta \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix} =$$

```
T at 20º:
     0.8830      0.1170      0.6428
     0.1170      0.8830     -0.6428
    -0.3214      0.3214      0.7660
```

MATLAB

$$[\bar{S}] = [T]^T[S][T]$$

$$\begin{Bmatrix} e_1 \\ e_2 \\ e_3 \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{12} & S_{22} & 0 \\ 0 & 0 & S_{33} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ 0 \end{Bmatrix}$$

$[\bar{S}] =$

```
S_bar_stress [1/MPa]:
     0.15641    -0.040438     -0.30099
    -0.040438     0.71266     -0.16576
     -0.30099    -0.16576      0.99764
```

$$e_1 = S_{11}\sigma_1 = \frac{1}{E_{11}}\sigma_1$$

$$E_{11} = \frac{1}{S_{11}} = \frac{1}{0.15641}$$

$$E_1 = 6.3933 \text{ MPa}$$

plane stress $\{\varepsilon\} = [Q]\{\sigma\}$ $\quad \nu_y = \frac{E_y \nu_x}{E_x} = \frac{1.3}{23.2}(0.28) = 0.01569$

$$[\bar{Q}] = \begin{bmatrix} \frac{E_x}{1-\nu_x\nu_y} & \frac{\nu_x E_y}{1-\nu_x\nu_y} & 0 \\ \frac{\nu_y E_x}{1-\nu_x\nu_y} & \frac{E_y}{1-\nu_x\nu_y} & 0 \\ 0 & 0 & G_{xy} \end{bmatrix} =$$

```
Q_bar [MPa]:
      18.635       2.4601       6.0309
      2.4601       1.7844       1.0387
      6.0309       1.0387       2.9945
```

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_6 \end{Bmatrix} = \begin{bmatrix} Q_1 & Q_{12} & 0 \\ Q_{12} & Q_2 & 0 \\ 0 & 0 & Q_3 \end{bmatrix} \begin{Bmatrix} e_1 \\ e_2 \\ e_3 \\ 0 \end{Bmatrix}$$

$$\sigma_1 = Q_1 e_1 = E_1^* e_1$$

$$E_1^* = 18.635 \text{ MPa}$$

```
clc
clear
close all
format short g
```

Problem 1

```
syms c s nu E

C_stress = E/(1 - nu^2) * [1 nu 0; nu 1 0; 0 0 (1-nu)/2];
T = [c*c s*s 2*c*s; s*s c*c -2*c*s; -s*c s*c c*c-s*s];
R = diag([1,1,2]);

C_prime_stress = T * C_stress * R / T / R;

diff_stress = C_stress - C_prime_stress;

disp("Plane stress C - C_prime diff:")
disp(simplify(diff_stress))


C_strain = E/(1 - nu^2) * [1 nu 0; nu 1 0; 0 0 (1-nu)/2];

C_prime_strain = T * C_strain * R / T / R;

diff_strain = C_strain - C_prime_strain;

disp("Plane strain C - C_prime diff:")
disp(simplify(diff_strain))
```

```
Plane stress C - C_prime diff:
[0, 0, 0]
[0, 0, 0]
[0, 0, 0]

Plane strain C - C_prime diff:
[0, 0, 0]
[0, 0, 0]
[0, 0, 0]
```

Problem 2

```
Ex = 23.2 * 1; % msi
Ey = 1.3 * 1; % msi
G = 0.90 * 1; % msi
nu_x = 0.28;
angle = deg2rad(20);

S = cast([1/Ex -nu_x/Ex 0; -nu_x/Ex, 1/Ey, 0; 0 0 1/G], "double");
T_20 = cast(subs(T, {c, s}, {cos(angle), sin(angle)}), 'double');
```

```
disp("S [1/MPa]:")
disp(S)
% fprintf(%X.Yf, S)
disp("T at 20º:")
disp(T_20)

S_bar = T_20' * S * T_20;

disp("S_bar_stress [1/MPa]:")
disp(S_bar)

E1 = 1/S_bar(1,1);
disp("E1 [MPa]:")
disp(E1)

nu_y = Ey / Ex * nu_x;
Q = cast([
    Ex/(1-nu_x*nu_y)      nu_x*Ey/(1-nu_x*nu_y) 0;
    nu_y*Ex/(1-nu_x*nu_y) Ey/(1-nu_x*nu_y) 0;
    0 0 G
    ], "double");
disp("Q [MPa]:")
disp(Q)

Q_bar = T_20 \ Q / T_20';

disp("Q_bar [MPa]:")
disp(Q_bar)

E1_star = Q_bar(1,1);
disp("E1_star [MPa]:")
disp(E1_star)
```

```
S [1/MPa]:
     0.043103    -0.012069            0
    -0.012069      0.76923            0
            0            0       1.1111

T at 20º:
      0.88302      0.11698      0.64279
      0.11698      0.88302     -0.64279
     -0.32139      0.32139      0.76604

S_bar_stress [1/MPa]:
      0.15641    -0.040438     -0.30099
    -0.040438      0.71266     -0.16576
     -0.30099     -0.16576      0.99764

E1 [MPa]:
       6.3933

Q [MPa]:
       23.302      0.36561            0
      0.36561       1.3057            0
```

```
            0            0          0.9

Q_bar [MPa]:
       18.635       2.4601       6.0309
       2.4601       1.7844       1.0387
       6.0309       1.0387       2.9945

E1_star [MPa]:
       18.635
```

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